

## 9-2g Revising Depreciation Estimates

Estimates of residual values and useful lives of fixed assets may change due to abnormal wear and tear or obsolescence. When new estimates are made by management, they are used to determine the depreciation expense in future periods. The depreciation expense recorded in earlier years is not affected.

To illustrate, assume the following data for a machine that was purchased on January 1:

|                                                                                                    |           |
|----------------------------------------------------------------------------------------------------|-----------|
| Initial machine cost                                                                               | \$140,000 |
| Expected useful life                                                                               | 5 years   |
| Estimated residual value                                                                           | \$ 10,000 |
| Annual depreciation using the straight-line method $[(\$140,000 - \$10,000) \div 5 \text{ years}]$ | \$ 26,000 |

At the end of the second year, the machine's book value (undepreciated cost) is \$88,000, computed as follows:

|                                                               |                  |
|---------------------------------------------------------------|------------------|
| Initial machine cost                                          | \$140,000        |
| Accumulated depreciation (\$26,000 per year $\times$ 2 years) | (52,000)         |
| Book value (undepreciated cost), end of second year           | <u>\$ 88,000</u> |

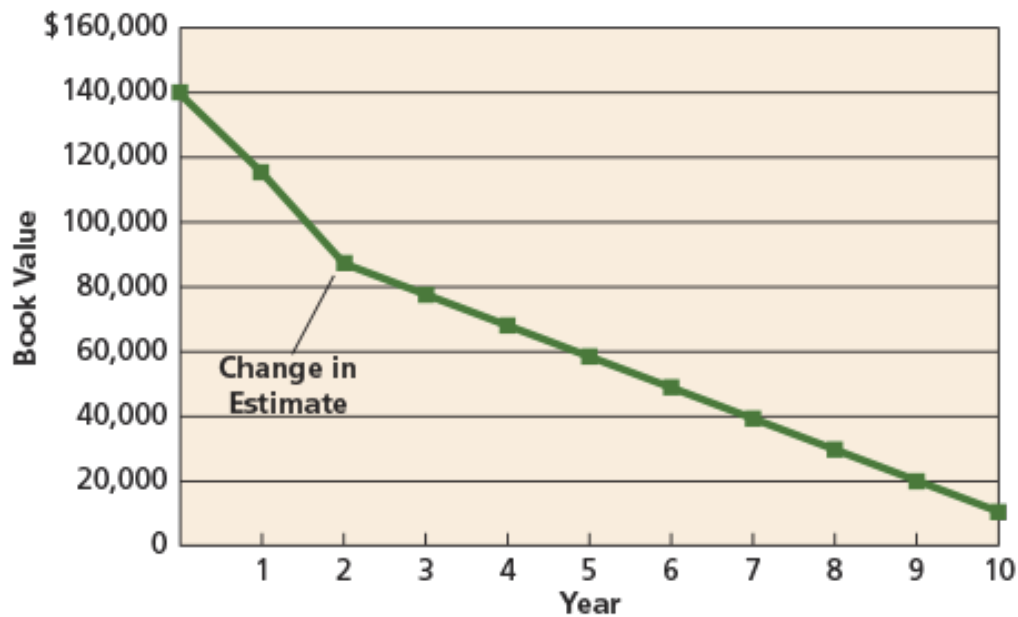
At the beginning of the third year, the company estimates that the machine's remaining useful life is eight years (instead of three) and that its residual value is \$8,000 (instead of \$10,000). The depreciation expense for each of the remaining eight years is \$10,000, computed as follows:

$$\text{Revised Depreciation Expense} = \frac{\text{Book Value} - \text{Revised Residual Value}}{\text{Revised Remaining Useful Life}} = \frac{\$88,000 - \$8,000}{8 \text{ Years}} = \$10,000$$

Exhibit 10 shows the book value of the asset over its original and revised lives. After the depreciation is revised at the end of the second year, book value declines at a slower rate. At the end of the tenth year, the book value reaches the revised residual value of \$8,000.

### Exhibit 10

#### Book Value of Asset with Change in Estimate



Using Data Analytics

### Maintaining Equipment

Maintaining equipment is a critical activity for capital-intensive companies like **United Parcel Services (UPS)**. As one of the world's largest package delivery companies, UPS operates over 500 aircraft and approximately 125,000 package cars, vans, trucks, and motorcycles. As a result, UPS employs over 4,000 aircraft and auto mechanics and spends millions of dollars on repairs and maintenance each year.

For UPS, scheduling preventive maintenance and avoiding equipment breakdowns is an integral part of its operations. Some of UPS's maintenance activities are mandated by regulatory authorities. For example, the Federal Aviation Administration (FAA) regulates UPS's maintenance and inspection of its aircraft. However, UPS has more discretion over scheduling and maintaining its 125,000 cars, vans, trucks, and motorcycles.

Data analytics can be useful to companies like UPS in scheduling maintenance and deciding when to replace its cars, vans, trucks, and motorcycles. For example, UPS could use data analytics to prepare an analysis of breakdowns and repairs costs by type of delivery vehicle. This might show that trucks break down more frequently and have more costly repairs than vans. This might result in UPS replacing some of its trucks with vans. Likewise, an analysis of breakdowns by geographic area might show more breakdowns for delivery vehicles in hot climates such as Arizona. This might result in UPS scheduling more frequent oil changes and tire replacements in these areas.

See [TIF 9-5](#) for a homework assignment using data analytics.

Chapter 9: Long-Term Assets: Fixed and Intangible: 9-2g Revising Depreciation Estimates

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